

Affine returns on Bernoulli trials in finance

Z.W.T. Mason

Stoch Answers Ltd, Sheffield, UK

Abstract

Offset returns of stock price movements were used to model the Paris Bourse in 1900. This first Mathematical model of Brownian motion was superseded by Geometric Brownian motion in the 1960s. i.e. the normal distribution was replaced by the log-normal and offset returns by linear returns. The market crash of 1998 caused the latter model to be questioned. This paper extends the model to affine returns, matching the average behaviour seen on the S&P 500 to calculations from averages of the daily ups and downs along with their probabilities. i.e. expected behaviour to noise. Affine returns lead to a mixture distribution consisting of two components — the log-normal distribution and one other which is fat-tailed.

1. Introduction

In his thesis Bachelier [1] derived a Mathematical model of Brownian Motion for the stock prices on the Paris Bourse, this was promptly ignored due to finance not being deemed a suitable topic for academic study. Five years later Einstein produced his model of Brownian Motion during his annus mirabilis. On the basis of a, possibly unfair, coin toss one set amount is won or a, possibly different, one is lost resulting in a binomial distribution which is approximated by a normal distribution. In the 1960s Mandelbrot [2] showed that cotton prices weren't normally distributed, similar studies for other assets followed. Geometric Brownian Motion was developed as a better model in which the stake grows or shrinks by set ratios leading to a log-normal distribution. This was used as the basis for deriving the Black-Scholes model for pricing European options. In 1997 Scholes and Merton were awarded the Nobel prize in economics, Black having died in the interim. At the time the two had left academia to work for Long Term Capital Management, the following year the US Federal Reserve intervened to save the market after the use of the model by their employer had racked up billions of dollars in losses.

This led to a re-examination of the foundations of quantitative finance. Wilmott [3] states that the returns on stocks progress from fat tailed to looking like they are normally distributed and finally to log-normal. These fat tails are what Taleb [4] calls Black Swans and not the long tails of the normal distribution which David Viniar, CFO of Goldman Sachs, apparently assumed despite their obvious impossibility:

We were seeing things that were 25 standard deviation moves, several days in a row.

In the abstract of his book Mandelbrot [5] dismisses models based on Bernoulli trials as mere coin tossing games:

Conventional financial theory assumes that variation of prices can be modeled by random processes that, in effect, follow the simplest “mild” pattern, as if each uptick or downtick were determined by the toss of a coin. What fractals show, and this book describes, is that by that standard, real prices “misbehave” very badly.

2. Affine returns on Bernoulli trials

Ignoring Mandelbrot and his propensity to see fractals everywhere in favour of Wilmott raises the question of how does the return progress from fat tailed to log-normal? The next level of complexity for a coin tossing game is to make the returns affine functions. i.e. With probability $q = 1 - p$ the payout for the next toss is a function of the stake returned at the t 'th toss, S_t ,

$$S_{t+1} = \alpha S_t + \gamma \quad (1)$$

and with probability p the payout is

$$S_{t+1} = \beta S_t + \delta \quad (2)$$

For $\alpha = \beta = 1$ the result is the binomial distribution for Brownian Motion which, by de Moivre-Laplace, tends to the normal distribution

$$S_t - S_0 \sim t\gamma + (\delta - \gamma) \mathcal{N}(tp, tpq) \quad (3)$$

hence

$$E[S_t] \sim S_0 + tq\gamma + tp\delta \sim E[S_{t-1}] + q\gamma + p\delta \quad (4)$$

For $\gamma = \delta = 0$, $0 < \alpha < 1 < \beta$ there is the binomial distribution for Geometric Brownian Motion which tends to the log-normal distribution

$$\log\left(\frac{S_t}{S_0}\right) \sim t \log(\alpha) + \log\left(\frac{\beta}{\alpha}\right) \mathcal{N}(tp, tpq) \quad (5)$$

hence

$$E[S_t] \sim S_0 \alpha^{tq} \beta^{tp} \sim E[S_{t-1}] \alpha^q \beta^p \quad (6)$$

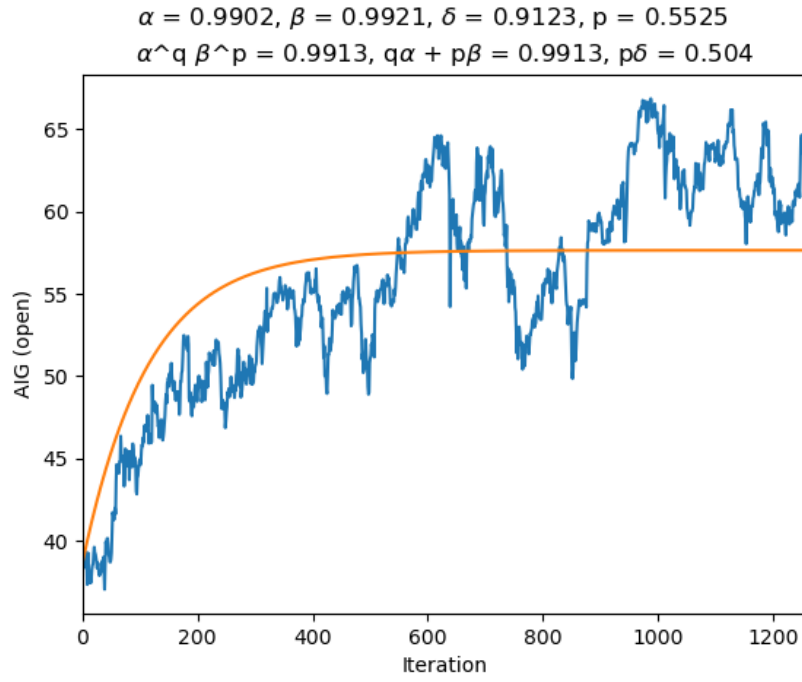
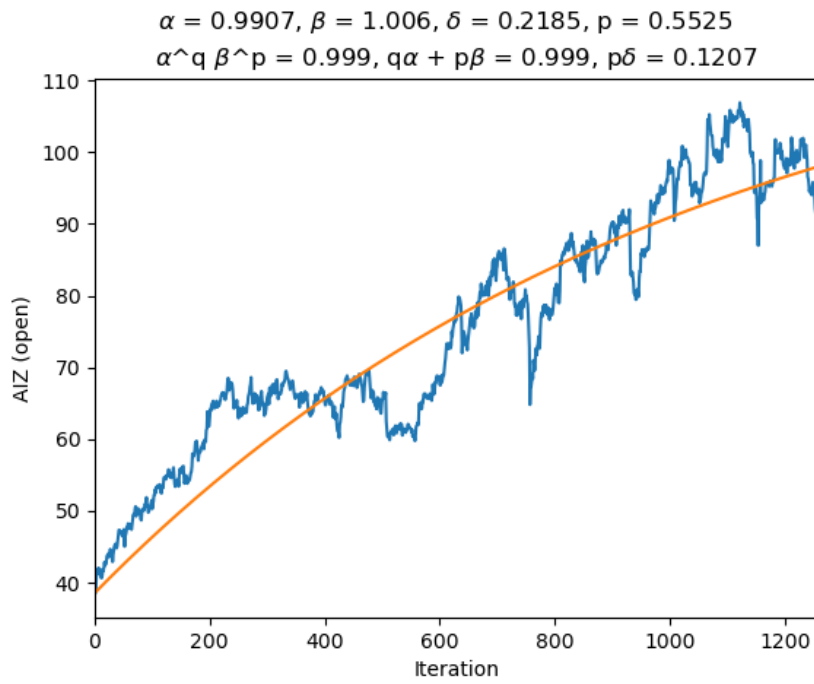
The more general case of $0 < \alpha < 1$, $0 < \beta \neq 1$, $0 = \gamma < \delta$ results in an order dependent set of terms known as a non-recombinant lattice. However, the terms containing the value of S_0 can be separated out showing that the result is a mixture distribution composed of the log-normal distribution and another. To see whether this is all just idle speculation the daily opening prices from the S&P 500 index for the 5 years from the 8th of February 2013 were analysed. The daily downs were fitted to equation 1, the daily ups to equation 2. Then the expected behaviour was approximated as

$$E[S_t] \sim (\alpha^q \beta^p)^t S_0 + \frac{r^t - 1}{r - 1} p\delta \text{ where } r = q\alpha + p\beta \neq \alpha^q \beta^p \quad (7)$$

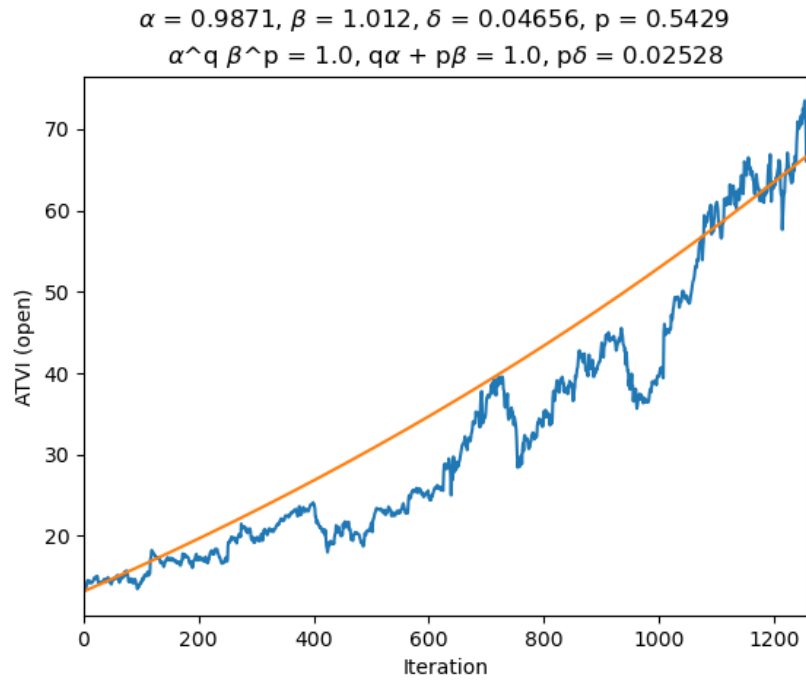
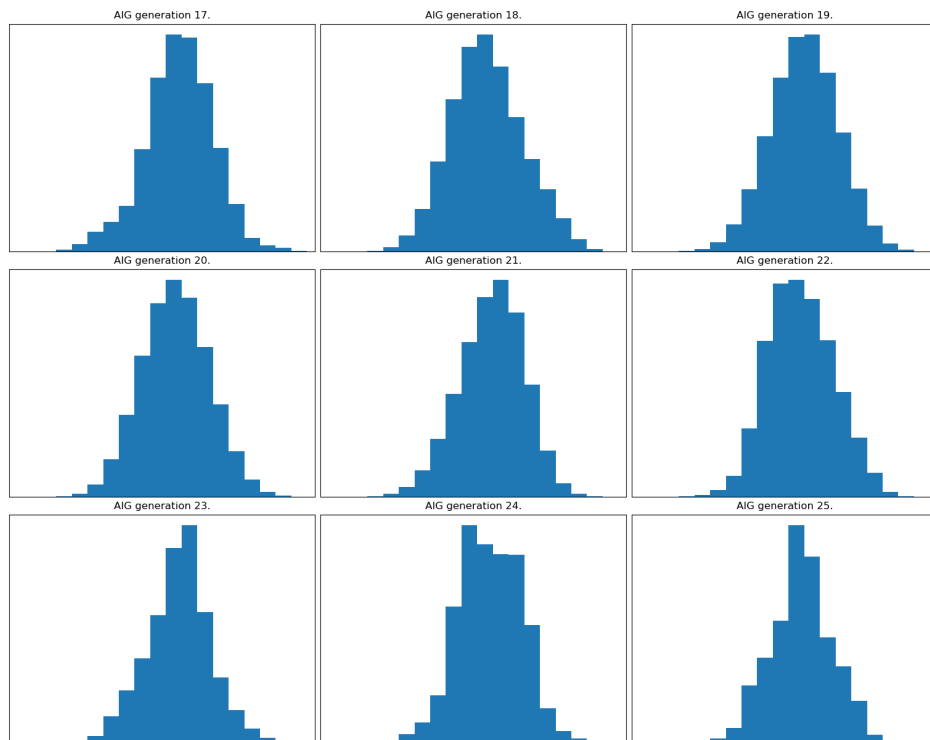
Note that, if $S_0 = 0$, $r = 1$, this compares to the recurrence relationship for the normal distribution – equation 4. The second term is a weighted average which doesn't compare directly to the first but, unlike the first term, for $\alpha, \beta < 1$, it doesn't shrink away to nothing. Hence the model is wrong which leaves the question of whether or not the formula is useful. The resulting growth of the formula values for AIG, AIZ and ATVI are shown in figures 1 to 3 corresponding to whether the support is finite or semi-infinite (β less than or greater than one) and whether there is exponential growth or decay (r less than or greater than one). These results were taken from a handful of decent fits however the correlations may, of course, be spurious. Nine generations of histograms of the mixture distribution are shown for the selected cases in figures 4 to 6 for the reader to consider whether or not they match the behaviour seen by Wilmott. The calculations were performed in C# using arbitrarily sized rational numbers to generate each possible point in a generation. Note that the reason why only 25 generations were calculated was because the algorithm is exponential in both time and storage.

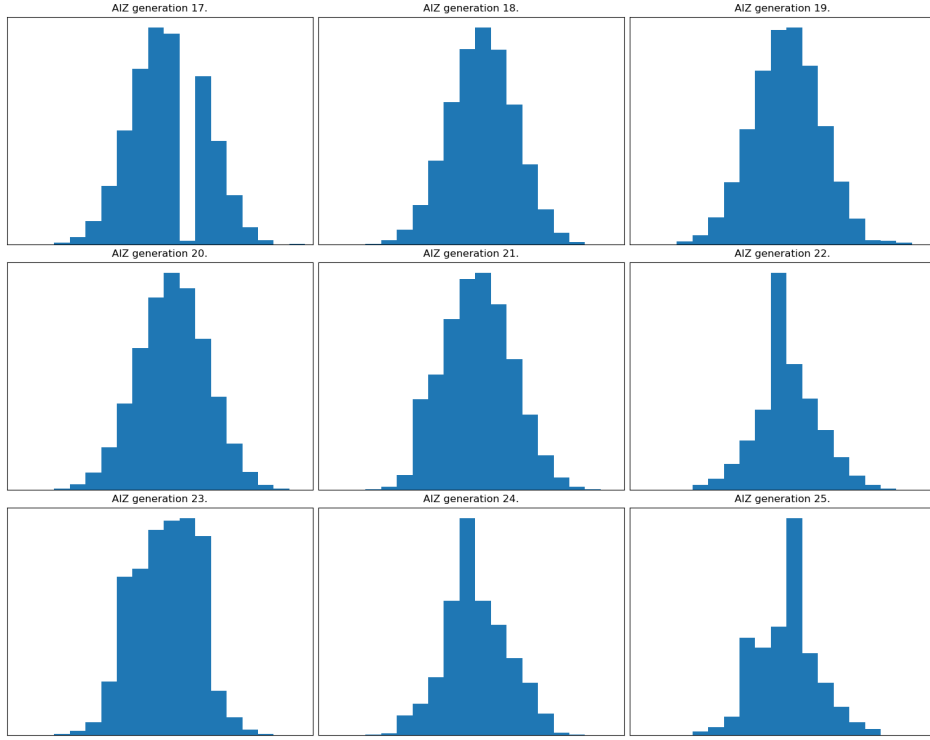
The second distribution is defined on the interval

$$\left(0, \delta \frac{1 - \beta^t}{1 - \beta}\right) \quad (8)$$

Figure 1. Fitting the noise for $\beta < 1$ Figure 2. Fitting the noise for $\beta > 1, q\alpha + p\beta < 1$

Defining a new random variable $X_t \in [0, 1]$ as S_t divided by the support when $S_0 = 0$ results in an expected value

Figure 3. Fitting the noise for $\beta > 1, q\alpha + p\beta > 1$ Figure 4. Mixture distributions by generation for $\beta < 1$

Figure 5. Mixture distributions by generation for $\beta > 1, q\alpha + p\beta < 1$

of

$$E[X_t] = p \frac{1-\beta}{1-r} \frac{1-r^t}{1-\beta^t} \quad (9)$$

which converges to some non-zero value for finite support, $\beta < 1$, and exponential decay, $r < 1$. In general, the moments can be expressed as a weighted sum where the weights sum to one.

$$E[X_t^n] = \sum_j w_j x_j^n \quad (10)$$

Applying the transform to individual points and weights

$$\{(w_j, x_j)\} \rightarrow \{(qw_j, \alpha x_j)\} \cup \{(pw_j, \beta x_j + 1 - \beta)\} \quad (11)$$

which implies at convergence that

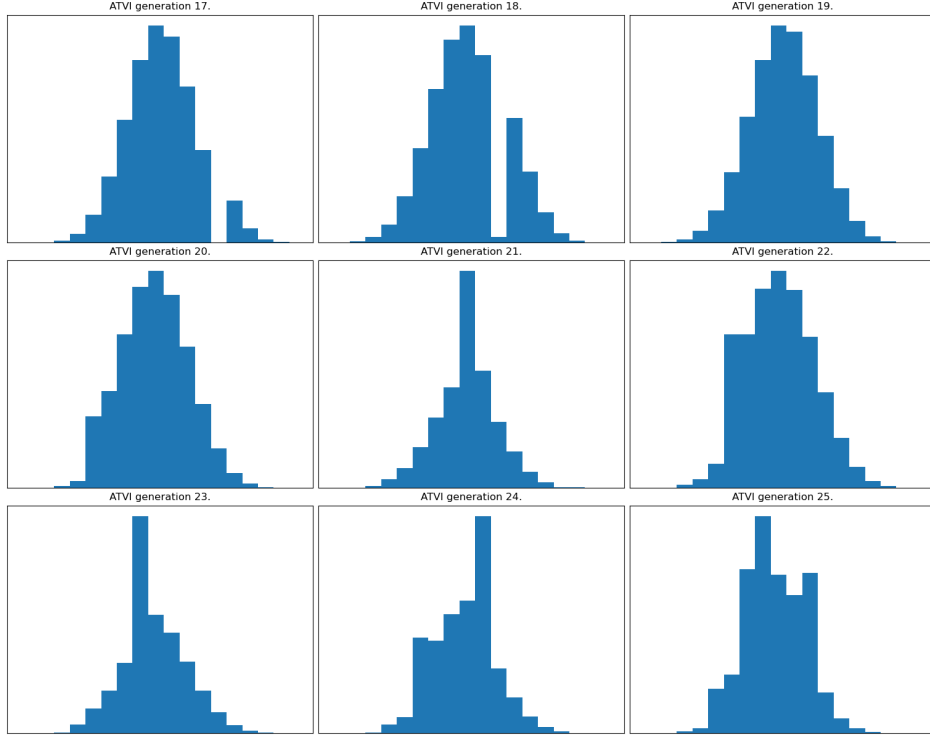
$$E[X^n] \rightarrow qE[(\alpha X)^n] + pE[(\beta X + 1 - \beta)^n] \quad (12)$$

and thus

$$E[X^n] = p \frac{E[(\beta X + 1 - \beta)^n] - \beta^n E[X^n]}{1 - q\alpha^n - p\beta^n} \quad (13)$$

The formulae for the first four moments are compared to calculations from the actual affine terms, equations 1 and 2, extrapolated as decaying exponentially to convergence in Appendix A.

Similarly, the transform can be applied to the probability distribution function. In the middle this is

Figure 6. Mixture distributions by generation for $\beta > 1, q\alpha + p\beta > 1$

$$f(x) \rightarrow qf\left(\frac{x}{\alpha}\right) + pf\left(1 - \frac{1-x}{\beta}\right) \quad (14)$$

Given that the transform involves a scale and a shift the transformed function at the tails is only due to a single instance rather than the sum of both. In particular, at the left tail when f is not constant

$$f(x) = qf(x/\alpha) \sim ax^m \Rightarrow m = \frac{\log q}{\log \alpha} \quad (15)$$

Similarly, the right tail is also fat

$$f(x) = pf\left(1 - \frac{1-x}{\beta}\right) \sim b(1-x)^n \Rightarrow n = \frac{\log p}{\log \beta} \quad (16)$$

For the special case of $\alpha = \beta \leq 1/2$ the ordering of the points is straightforward.

$$x_i = \delta \sum_{j=0}^n a_j \beta^j, \quad a_j \in \{0, 1\} \quad (17)$$

where i has the binary expansion $a_0 \cdots a_n$ and the weights are

$$w_i = q^{n-m} p^m, \quad m = \sum_{j=0}^n a_j \quad (18)$$

3. The Dirac delta distribution

For the case $\alpha = \beta$ equation 13 produces

$$E[X^n] = p \frac{E[(\beta X + 1 - \beta)^n] - \beta^n E[X^n]}{1 - \beta^n} = p \frac{n\beta^{n-1} E[X^{n-1}] (1 - \beta) + O((1 - \beta)^2)}{1 - \beta^n} \quad (19)$$

hence as $\beta \rightarrow 1$

$$E[X^n] \rightarrow p E[X^{n-1}] = p^n = \int_0^1 x^n \delta(x - p) dx \quad (20)$$

4. The uniform distribution

For the case $\alpha = \beta = p = q = 1/2$ the numerical calculations appear to show a uniform distribution. Substituting those values into equation 13 produces

$$E[X^n] = \frac{E[(X + 1)^n] - E[X^n]}{2^{n+1} - 2} \quad (21)$$

which is verified by substituting the moments of the uniform distribution. i.e.

$$E[X^n] = \int_0^1 x^n dx = \frac{1}{n+1} \quad (22)$$

and

$$E[(X + 1)^n] = \int_0^1 (x + 1)^n dx = \int_1^2 x^n dx = \frac{1}{n+1} (2^{n+1} - 1) \quad (23)$$

5. Discussion

It was shown that it is better to use a mixture distribution (due to affine returns) rather than merely the log-normal distribution (due to linear returns). This results in a model for the stock price of Geometric Brownian Motion plus an extra term

$$S_t = S_0 \alpha^t \left(\frac{\beta}{\alpha} \right)^{t p + \sqrt{p q} W_t} + \delta \left(\frac{1 - \beta^t}{1 - \beta} \right) X(p, \alpha, \beta, t, W_t) \quad (24)$$

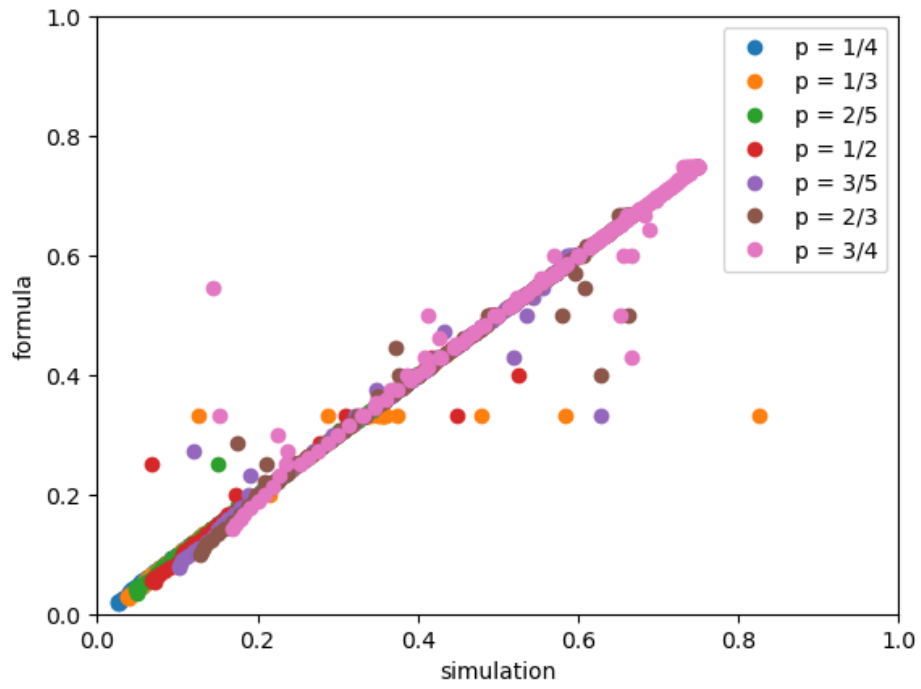
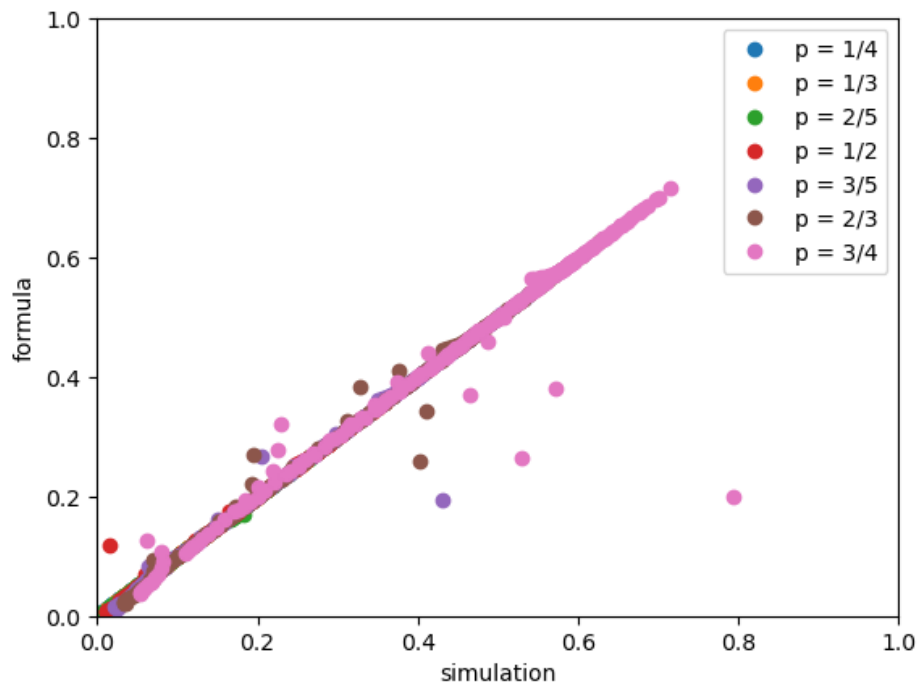
where X is taken from a probability distribution with finite support (over $[0, 1]$) that is fat-tailed.

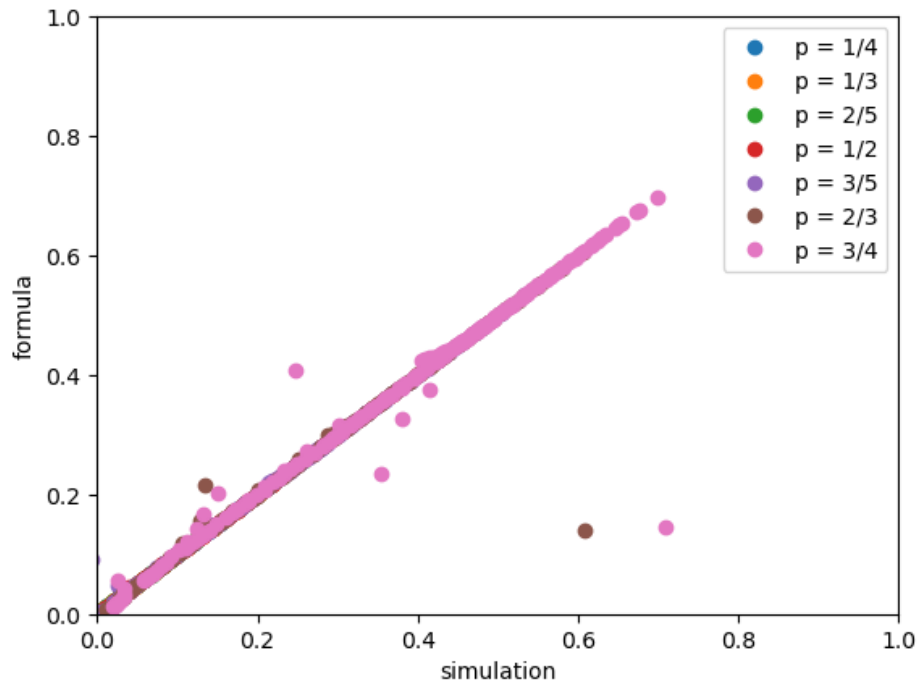
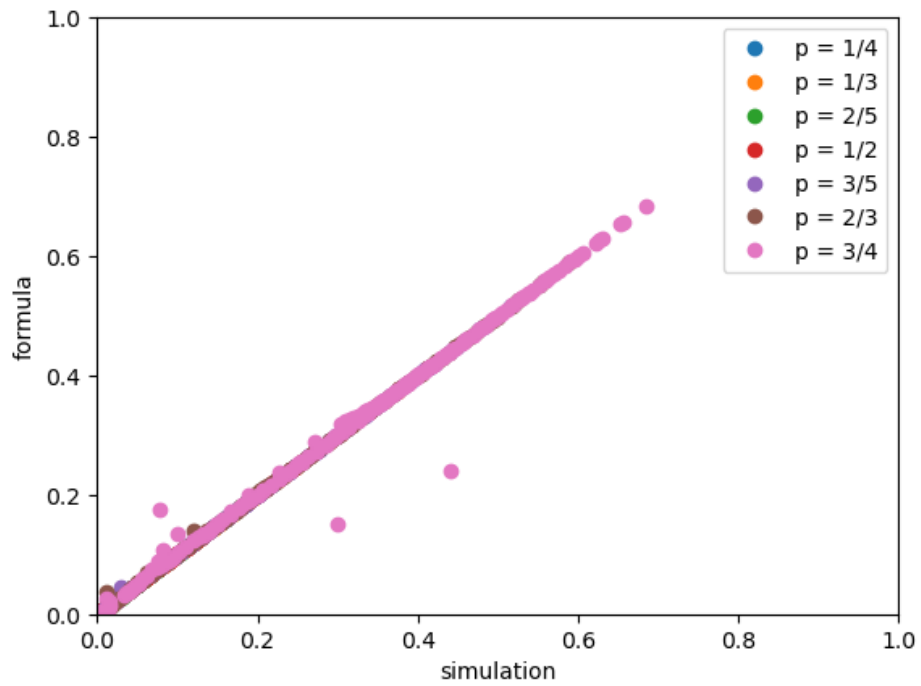
For the case of $\alpha, \beta < 1$, as verified by simulated data, the moments are known. Even so the probability density function may not even exist or, if it does, then may not be uniquely determined according to Mnatsakanov and Hakobyan [6]. It is possible to construct a set of polynomial basis functions which are all zero at the end points by using Gegenbauer polynomials with a parameter of $5/2$ and evaluate their coefficients from the functions for the moments. Any approximation to the probability density function built using these polynomials is expected to be wobbly giving rise to negative values.

Rather than guess a continuous model there is a discrete model at hand given by equations 1 and 2. This can be used to construct a non-recombinant binomial options pricing model. Such pricing models have the same issue as was seen in calculating the discrete distribution, namely that they are exponential in both time and storage. As has been mentioned for the authors 16GB machine the depth of the binary tree was limited to 25 generations. As such memory limitations may mean an implementation will not be sufficient to be useful.

Appendix A. Moments

The first four moments are compared to values calculated from the probability distributions in figures A.7 to A.10.

Figure A.7. Comparison of calculations to formula for $E(X)$ Figure A.8. Comparison of calculations to formula for $E(X^2)$

Figure A.9. Comparison of calculations to formula for $E(X^3)$ Figure A.10. Comparison of calculations to formula for $E(X^4)$

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